Valuation of Bonds and Stocks

Types, Ratings, Interest Rates, and Dividend Models

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1. Types and Features of Bonds

A bond is a debt instrument where the issuer owes the holders a debt and is obliged to pay interest (coupons) and/or repay the principal.

- **Common Types:**
  - **Treasury Bonds:** Issued by federal governments (low default risk).
  - **Corporate Bonds:** Issued by firms; varying levels of risk.
  - **Municipal Bonds:** Issued by local governments (often tax-exempt).
  - **Zero-Coupon Bonds:** No periodic interest; sold at a deep discount.
- **Key Features:**
  - **Par Value:** The face value (usually $1,000).
  - **Coupon Rate:** Stated interest rate.
  - **Maturity:** Time until the principal is repaid.
  - **Call Provision:** Allows issuer to repurchase bond before maturity.

2. Price vs. Intrinsic Value and Fluctuations

- **Intrinsic Value:** The present value (PV) of all expected future cash flows, discounted at the required rate of return (r).

$$V_B = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{M}{(1+r)^n}$$

- **Price vs. Value:** The market price converges to the intrinsic value in efficient markets.
- **Why do prices fluctuate?**
  - **Inverse Relationship:** When market interest rates (r) rise, bond prices fall. When r falls, prices rise.
  - **Interest Rate Risk:** Longer-term bonds are more sensitive to rate changes than short-term bonds.

3. Bond Ratings and Their Meaning

Ratings assess the **creditworthiness** of the issuer (the likelihood of default).

Category	**S&P / Fitch**	**Moody's**
Investment Grade	AAA, AA, A, BBB	Aaa, Aa, A, Baa
Speculative (Junk)	BB, B, CCC, D	Ba, B, Caa, C

- **High Rating:** Lower default risk $\rightarrow$ Lower yield required.
- **Low Rating:** Higher default risk $\rightarrow$ Higher yield required (Risk Premium).
- **Determinants:** Financial ratios (D/E, Interest Coverage), business stability, and economic environment.

Credit Rating Scales Categories

Moody's	S&P	Fitch Ratings	Pefindo	Kategori
Aaa	AAA	AAA	idAAA	*Investment Grade*
Aa1	AA+	AA+	idAA+	*Investment Grade*
Aa2	AA	AA	idAA	*Investment Grade*
Aa3	AA–	AA–	idAA–	*Investment Grade*
A1	A+	A+	idA+	*Investment Grade*
A2	A	A	idA-	*Investment Grade*
A3	A–	A–	idA	*Investment Grade*
Baa1	BBB+	BBB+	idBBB+	*Investment Grade*
Baa2	BBB	BBB	idBBB	*Investment Grade*
Baa3	BBB–	BBB–	idBBB–	*Investment Grade*
Ba1	BB+	BB+	idBB+	*Non-Investment Grade*
Ba2	BB	BB	idBB	*Non-Investment Grade*
Ba3	BB–	BB–	idBB–	*Non-Investment Grade*
B1	B+	B+	idB+	*Non-Investment Grade*
B2	B	B	idB	*Non-Investment Grade*
B3	B–	B–	idB-	*Non-Investment Grade*
Caa	CCC	CCC	idCCC	*Non-Investment Grade*
Ca	CC	CC	idSD	*Non-Investment Grade*
C	C	C	idD	*Non-Investment Grade*
–	D	D	–	*Default*

4. Impact of Inflation on Interest Rates

Investors demand compensation for the loss of purchasing power.

- **The Fisher Effect:** Describes the relationship between nominal rates (R), real rates (r^*), and expected inflation (h).

$$(1 + R) = (1 + r^*)(1 + h)$$

- **Approximation:** $R \approx r^* + h$.
- **Impact:** If expected inflation rises, lenders demand higher nominal interest rates to maintain their real return, causing bond prices to drop.

5. Term Structure and Bond Yields

The **Term Structure** is the relationship between interest rates (or yields) and maturities for default-free bonds.

- **The Yield Curve:** Graph of this relationship (Normal, Inverted, or Flat).
- **Factors Affecting Yields:**
  1. **Real Risk-Free Rate** (r^*)
  2. **Inflation Premium** (IP)
  3. **Maturity Risk Premium** (MRP) - risk of price fluctuations.
  4. **Default Risk Premium** (DRP)
  5. **Liquidity Premium** (LP)

$$\text{Yield} = r^* + IP + MRP + DRP + LP$$

6. YTM Calculation: The Approximation Method

The Yield to Maturity (YTM) is the expected return if the bond is held until it matures. A quick estimate can be found using the **Approximation Formula**:

Approximation Formula

$$YTM \approx \frac{C + \frac{F-P}{n}}{\frac{F+P}{2}}$$

Example: A 10-year bond, 8% annual coupon ($80), Par value ($1,000), currently selling for $920.

- $C = 80, F = 1000, P = 920, n = 10$
- $YTM \approx \frac{80+\frac{1000-920}{10}}{\frac{1000+920}{2}} = \frac{80+8}{960} = \frac{88}{960} \approx \mathbf{9.17\%}$

7. YTM Calculation: Linear Interpolation

For more accuracy without a calculator, we use **Interpolation** between two trial discount rates (r_1 and r_2).

1. Select two rates where one price is above and one is below the market price (P).
2. **Formula:**

$$YTM = r_1 + \left(\frac{V_1 - P}{V_1 - V_2}\right) \times (r_2 - r_1)$$

Example (using the \$920 bond from before):

- Try $r_1 = 9\% \rightarrow V_1 = \$935.82$ (Above \$920)
- Try $r_2 = 10\% \rightarrow V_2 = \$877.07$ (Below \$920)
- $YTM = 0.09 + \left(\frac{935.82-920}{935.82-877.07}\right) \times (0.10 - 0.09)$
- $YTM = 0.09 + (0.269) \times 0.01 = \mathbf{9.269}\%$

8. Dividends and Stock Valuation

A stock's value is derived from the present value of all future dividends.

- **Dividend Payments:** Immediate cash flow to shareholders. Higher dividends (ceteris paribus) lead to higher stock prices.
- **Growth Rate (g):** The rate at which dividends are expected to increase.
- **The Trade-off:** Firms can pay out dividends now or reinvest earnings to increase g.
- **Impact:** If the growth rate g increases, the stock price increases because future cash flows will be larger.

9. Stock Valuation: The Gordon Growth Model

For a stock with dividends growing at a constant rate g forever:

The Formula

$$P_0 = \frac{D_1}{r_s - g} = \frac{D_0(1+g)}{r_s - g}$$

Where:

- P_0 = Current intrinsic value of the stock.
- D_1 = Dividend expected next period.
- r_s = Required rate of return.
- g = Constant growth rate.

Constraint: This model is only valid if $r_s > g$.

Example: Constant Dividend Growth Model

Scenario: Global Energy Corp just paid an annual dividend of **\$2.00** per share. The company is expected to increase its dividend by **6%** per year indefinitely. Investors require a **13%** return on this stock.

Step 1: Identify the variables

- $D_0 = \$2.00$ (The dividend just paid)
- $g = 6\% = 0.06$
- $r_s = 13\% = 0.13$

Step 2: Calculate the expected dividend (D_1)

$$D_1 = D_0 \times (1 + g) = 2.00 \times (1.06) = \$2.12$$

Step 3: Calculate the Intrinsic Value (P_0)

$$P_0 = \frac{D_1}{r_s - g} = \frac{2.12}{0.13 - 0.06} = \frac{2.12}{0.07} = \mathbf{\$30.29}$$

Sensitivity Analysis: What changes the price?

Using the formula $P_0 = \frac{D_1}{r_s - g}$, we can observe:

- **Risk Appetite:** If investors perceive more risk, r_s increases $\rightarrow$ **Stock Price Falls**.
- **Growth Expectations:** If the company finds better projects and g increases $\rightarrow$ **Stock Price Rises**.
- **The Gap:** As g approaches r_s, the stock price becomes extremely sensitive to small changes in assumptions (the "denominator effect").

Warning

If $g \geq r_s$, the formula yields a negative or infinite value, which is mathematically valid but economically impossible for a constant growth firm.

Practice 1: Semiannual Bond Pricing

Problem: A 15-year corporate bond has a par value of \$1,000 and a 7% annual coupon rate. If the required market rate of return is 9% and coupons are paid **semiannually**, calculate the bond price.

Solution

- $n = 15 \times 2 = 30$ periods; $r = 9\%/2 = 4.5\%$ per period
- $PMT = (7\% \times 1000)/2 = \$35$
- $Price = 35 \times \left[\frac{1-(1.045)^{-30}}{0.045}\right] + \frac{1000}{(1.045)^{30}}$
- $Price = 35 \times (16.2889) + 1000 \times (0.2670) = \$\mathbf{837.11}$

Practice 2: YTM Approximation

Problem: A 10-year bond with a 10% annual coupon rate and a par value of \$1,000 is currently selling for \$1,150. Calculate the approximate Yield to Maturity (YTM).

Solution

$$YTM \approx \frac{C + \frac{F-P}{n}}{\frac{F+P}{2}} = \frac{100 + \frac{1000-1150}{10}}{\frac{1000+1150}{2}}$$

$$YTM \approx \frac{100 - 15}{1075} = \frac{85}{1075} = \mathbf{7.91\%}$$

Practice 3: Zero-Coupon Valuation

Problem: You want to buy a 20-year zero-coupon bond ($1,000 face value). If your required return is 6% (compounded **semiannually**), what is the maximum price you should pay?

Solution

- $n = 20 \times 2 = 40$ periods
- $r = 6\%/2 = 3\% = 0.03$
- $Price = \frac{FV}{(1+r)^n} = \frac{1000}{(1.03)^{40}}$
- $Price = \frac{1000}{3.26204} = \$\mathbf{306.56}$

Practice 4: The Fisher Effect (Real Returns)

Problem: A 1-year T-bill offers a 5.5% nominal rate. If expected inflation is 2.8%, calculate the exact real rate of return.

Solution

$$(1+R) = (1+r^*)(1+h) \implies r^* = \frac{1+R}{1+h} - 1$$

$$r^* = \frac{1.055}{1.028} - 1 = 1.02626 - 1$$

$$r^* = 0.02626 \text{ or } \mathbf{2.63}\%$$

Practice 5: Determinants of Interest Rates

Problem: A corporate bond yields 8.5%. Given: $r^* = 2\%$, Inflation Premium = 3%, Liquidity Premium = 0.5%, and Maturity Risk Premium = 1%. Solve for the **Default Risk Premium (DRP)**.

Solution

$$NominalYield = r^* + IP + MRP + DRP + LP$$

$$8.5\% = 2\% + 3\% + 1\% + DRP + 0.5\%$$

$$DRP = 8.5\% - 6.5\% = \mathbf{2.0}\%$$

Practice 6: Constant Growth Stock

Problem: A stock just paid a dividend (D_0) of \$2.50. It is expected to grow at 5% indefinitely. If the required return is 12%, find the intrinsic value.

Solution

- $D_1 = D_0(1+g) = 2.50(1.05) = \$2.625$
- $P_0 = \frac{D_1}{r_s - g} = \frac{2.625}{0.12 - 0.05}$
- $P_0 = \frac{2.625}{0.07} = \mathbf{\$37.50}$

Practice 7: Finding the Required Return

Problem: Stock price is \$45.00. Next expected dividend (D_1) is \$3.15. Growth rate is 4%. What is the required rate of return (r_s)?

Solution

$$r_s = \frac{D_1}{P_0} + g$$

$$r_s = \frac{3.15}{45.00} + 0.04$$

$$r_s = 0.07 + 0.04 = 0.11 \text{ or } \mathbf{11\%}$$

Practice 8: ROE and Growth

Problem: A firm has an ROE of 15% and a payout ratio of 40%. Most recent dividend (D_0) was \$2.00. Required return is 11%. Calculate g and the stock price.

Solution

- $g = (1 - \text{Payout}) \times ROE = (1 - 0.40) \times 0.15 = \mathbf{9\%}$
- $P_0 = \frac{2.00(1.09)}{0.11-0.09} = \frac{2.18}{0.02} = \mathbf{\$109.00}$

Practice 9: Dividend vs. Capital Gains Yield

Problem: Using $D_0 = \$2.50$, $g = 5\%$, and $P_0 = \$37.50$, find the Dividend Yield and Capital Gains Yield for Year 1.

Solution

- **Div. Yield** $= \frac{D_1}{P_0} = \frac{2.50(1.05)}{37.50} = \mathbf{7\%}$
- **Cap. Gains Yield** $= \frac{P_1 - P_0}{P_0} = g = \mathbf{5\%}$
- Total Return $= 7\% + 5\% = 12\%$ (Matches r_s)

Practice 10: Premium Bond Amortization

Problem: A 5-year bond (8% coupon) is bought for \$1,085. If YTM is 6%, what will the price be exactly 1 year from now?

Solution

- After 1 year, $n = 4$ years remaining.
- $P_1 = 80 \times \left[\frac{1-(1.06)^{-4}}{0.06}\right] + \frac{1000}{(1.06)^4}$
- $P_1 = 80 \times (3.4651) + 1000 \times (0.7921) = \mathbf{\$1,069.31}$
- (Note: The premium "wastes away" as it nears maturity)

Thank You!

Q&A Session